EXCEL

Excel for Data Analysts

The functions that cover daily analyst work, plus the 2 debugging questions every interviewer asks.

- **8 functions with real examples**
- **The #N/A checklist**
- **Why SUM shows 0**

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Lookup and summarise

This is 70% of Excel work in any analyst job.

XLOOKUP

Looks left or right, handles "not found" on its own. Replaces VLOOKUP.

```
=XLOOKUP(A2, Emp[ID], Emp[Name], "Not found")
```

SUMIFS

Sum with any number of conditions.

```
=SUMIFS(Sales, Region, "North", Month, "Mar")
```

COUNTIFS

Count with conditions. Great for open tickets per owner.

```
=COUNTIFS(Status, "Open", Owner, G2)
```

IFS

Clean buckets without nesting 5 IFs.

```
=IFS(B2>=90,"A", B2>=75,"B", TRUE,"C")
```

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Clean and shape

Raw data is never clean. These save hours every week.

TRIM + CLEAN

Removes extra and invisible spaces that break lookups.

```
=TRIM(CLEAN(A2))
```

TEXTSPLIT

Split "Delhi, North, 2026" into 3 columns.

```
=TEXTSPLIT(A2, ", ")
```

UNIQUE + FILTER

Live lists that update when data changes.

```
=FILTER(Data, Data[Sales] > 50000)
```

TEXT

Turn dates into readable labels for reports.

```
=TEXT(A2, "mmm-yy")
```

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Pivot table in 5 steps

The fastest way to summarise thousands of rows. Asked in almost every Excel round.

Raw rows

North · Jan · 120

South · Jan · 90

North · Feb · 140

South · Feb · 75

North · Jan · 60

→

Pivot

	Jan	Feb
North	180	140
South	90	75
Total	270	215

1. Ctrl + T

Turn data into a Table so the pivot grows with new rows.

2. Insert → PivotTable

Place it on a new sheet.

3. Rows and Values

Region to Rows, Sales to Values. Month to Columns.

4. Group dates

Right-click a date → Group → Months.

5. Show as %

Value Field Settings → Show Values As → % of Grand Total.

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VLOOKUP shows #N/A

Interviewers ask this to see if you have worked with real data. Check in this order.

1. Hidden spaces

"A101 " and "A101" do not match. Wrap the lookup value.

```
=VLOOKUP(TRIM(A2), Range, 2, FALSE)
```

2. Number stored as text

One side is 101, the other is "101". Green triangle in the corner gives it away. Fix with VALUE or Text to Columns.

3. Range not locked

Range shifts when copied down. Use $A$2:$C$500 or a Table.

4. Missing FALSE

Without it Excel does an approximate match.

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SUM shows 0

The numbers are there, the total is 0. Three causes, in order of how often I see them.

Numbers stored as text

Left aligned, green triangle. Select the column, Data, Text to Columns, Finish.

Calculation is Manual

Someone switched it to speed up a big file. Formulas, Calculation Options, Automatic.

Circular reference

The SUM range includes its own cell. Status bar shows the cell address.

Interview tip: say the checks in order. It shows you have debugged real files, not just practised clean data.

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Shortcuts that save hours

Speed is noticed in live Excel tests. Practise these until they are habit.

- **Ctrl + T**
  Convert range to a Table
- **Ctrl + Shift + L**
  Turn filters on or off
- **Alt + =**
  AutoSum the column above
- **Ctrl + Arrow keys**
  Jump to the edge of the data
- **F4**
  Lock a cell reference ($A$1) while writing a formula
- **Ctrl + ;**
  Insert today's date
- **Ctrl + Alt + V**
  Paste Special (values, formats, transpose)
- **Ctrl + 1**
  Format cells

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15 practice questions

Use a sales dataset with date, region, product, qty and price.

1. ☐ Lookup a value that sits left of the key column
2. ☐ Total North sales in March only
3. ☐ Flag orders above average order value
4. ☐ Extract the domain from an email
5. ☐ Count unique customers
6. ☐ Rank salespeople within each region
7. ☐ Convert 20261105 into a real date
8. ☐ Find duplicates across two sheets
9. ☐ Month-end date for every transaction
10. ☐ Create Low / Medium / High buckets
11. ☐ Auto-updating top 5 products by revenue
12. ☐ % contribution of each region
13. ☐ Working days between order and delivery
14. ☐ Combine first and last name
15. ☐ Dropdown that filters a table

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